VARIABLES AND FORMULAS

n: SAMPLE SIZE (# UNITS IN THE SAMPLE)

N: POPULATION SIZE

X: RANDOM VARIABLE DESCRIBED BY SOME PROBABILITY DISTRIBUTION

$\underline{x}$: A RANDOM SAMPLE $(x_1, x_2, \dots x_n)$

$\bar{x}$: SAMPLE MEAN

μ: POPULATION MEAN

σ^2: POPULATION VARIANCE

A SAMPLE STATISTIC IS A FUNCTION OF THE RANDOM SAMPLE $y = t(\underline{x}) = t(x_1, x_2, \dots, x_n)$, THE MOST COMMON STATISTICS ARE:

- SAMPLE MEAN $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- SAMPLE VARIANCE $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$
- SAMPLE STANDARD DEVIATION $s = \sqrt{s^2}$
- NON-CENTRAL MOMENT OF ORDER m $\quad m_r = \frac{1}{n}\sum_{i=1}^{n} x_i^r$
- CENTRAL MOMENT OF ORDER r $\quad \overline{m_r} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^r$

- $E(-2X) = -2E(X)$
- $VAR(s^2) = \frac{2\sigma^4}{n-1}$?

- $\mu = E(X) = E(\bar{x})$
- $\sigma^2 = VAR(X)$
- $\frac{\sigma^2}{n} = VAR(\bar{x})$
- $E(s^2) = \sigma^2$; $E(s) \leq \sigma$
- $VAR(y) = E(y^2) - E(y)^2$
- $VAR(s^2) = \frac{2\sigma^2}{n-1}$

- MARKOV INEQUALITY: $\begin{cases} P(X \geq \omega) \leq \frac{E(X)}{\omega} \\ P(X \leq \omega) \geq 1 - \frac{E(X)}{\omega} \end{cases}$
- CEBISHEV INEQUALITY: $\begin{cases} P(|X - E(X)| \leq \omega) \geq 1 - \frac{VAR(X)}{\omega^2} \\ P(|X - E(X)| \geq \omega) \leq \frac{VAR(X)}{\omega^2} \end{cases}$
- CLT: $n \geq 30$

9. METHODS OF EVALUATING ESTIMATORS

CHOOSE THE BEST ESTIMATOR OF θ ALONG ALL THE POSSIBLE ESTIMATORS OF THIS PARAMETER

- MEAN SQUARED ERROR (MSE)

BIAS

$B_\theta(T) = E_\theta(T) - \theta$

$MSE_\theta(T) = \underbrace{E_\theta\left[(T-\theta)^2\right]}_{\text{ESTIMATOR TO EVALUATE}} + V(T) = B_\theta(T)^2 + V(T)$

T^* IS THE BEST ESTIMATOR FOR θ IF $\forall \theta \in \Theta$ $MSE_\theta(T^*) \leq MSE_\theta(T)$

- NO NAME

$E_\theta(|T - \theta|)$

EXPONENTIAL FAMILY

A DISTRIBUTION BELONGS TO AN EXP. FAMILY IF

1) SUPP(X) DOES NOT DEPENDS ON θ ← IF YOU CHANGE THE PARAM. θ THE DOMAIN WILL NOT CHANGE
2) $f(x, \theta) = h(x)c(\theta)\exp\{w_1(\theta)t_1(x)\}$

IF THERE IS ANOTHER PARAM. IN THE EXPONENTIAL GIVE IT TO $t(x)$ AND LEAVE $w(\theta)$ ALONE

- NATURAL PARAMETRIZATION

$f(x;\theta) = h(x)c(\theta)\exp\{w_1(\theta)t_1(x)\} \Rightarrow f(x;\theta) = h(x)c^*(\eta)\exp\{\eta t(x)\}$

$\eta = w_1(\theta) \longrightarrow$ ISOLATE $\theta \longrightarrow$ REWRITE $c(\theta)$ WITH THE NEW θ (THAT DEPENDS ON η)

- EXPECTATION

$E(t(x)) = -\frac{d}{d\eta}\ln(c^*(\eta))$

STANDARIZED SAMPLE MEAN

$Z = \frac{\bar{x} - \mu}{\sqrt{\sigma^2/n}} \sim N(0,1) \xrightarrow{\text{IF UNKNOWN } \sigma^2} T = \frac{\bar{x} - \mu}{\sqrt{s^2/n}} \sim T_{n-1}$

ADD STANDARIZED SAMPLE VARIANCE

8. POINT ESTIMATION METHOD

→ REMEMBER TO ALWAYS PUT A ^ (HAT) BECAUSE ARE PREDICTED

- ESTIMATION BY ANALOGY: GENERALIZE FROM THE SAMPLE TO THE DISTRIBUTION

PREDICTED → $\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ (PREDICTED POPULATION MEAN = SAMPLE MEAN)

PREDICTED → $\hat{\sigma}^2 = s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ (PREDICTED POPULATION VARIANCE = SAMPLE VARIANCE)

↳ BIASED

- METHOD OF MOMENTS

$\begin{cases} m_1 = \mu_1 \\ m_2 = \mu_2 \\ \dots \end{cases} \Rightarrow \begin{cases} \frac{1}{n}\sum_{i=1}^{n} x_i = E(x) \\ \frac{1}{n}\sum_{i=1}^{n} x_i^2 = E(x^2) \\ \dots \end{cases}$ TO FIND

- MAXIMUM LIKELIHOOD ESTIMATOR
  - WRITE $f(x, \theta) \longrightarrow$ ESTIMATE θ
  - $\ln(f(x, \theta))$
  - $l(P) = \sum_{i=1}^{n}\ln(f(x_i, \theta)) \longrightarrow$ SUM WHAT YOU ARE NOT INTEREST TO
  - $\frac{d}{d\theta} l(P) = 0 \longrightarrow$ COMPUTE DERIVITATIVE WRT WHAT YOU ARE INTEREST TO $\longrightarrow$ PUT $= 0$ TO FIND THE MAXIMUM
  - ISOLATE θ

↗ NUMERICAL ALGORITHMS

↳ NEWTON-RAPHSON

FISHER SCORING

CONFIDENCE INTERVALS

- ↑% ⟷ WIDER C.I.
- ↓% ⟶ THINNER C.I.
- THE GIVEN % IS $1-\alpha$
- LENGTH $= L_2 - L_1$

PIVOTAL?

1. NORMAL POPULATION, KNOW POP. VARIANCE AND CI FOR THE POP. MEAN

- $z_{\alpha/2} = \text{qnorm}(1-\alpha/2)$

$$CI: \bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

- $\text{LENGTH}(CI) = L_2 - L_1 = 2z\frac{\sigma}{\sqrt{n}}$

3. GENERIC POPULATION, LARGE SAMPLE AND FIND CI FOR THE POP. MEAN

APPLY IF

- $n \geq 30$ W/ VARIANCE KNOWN
- $n \geq 30$ W/ VARIANCE UNKNOWN

] → CLT (CENTRAL LIMIT THEOREM)

- $z_{\alpha/2} = \text{qnorm}(1-\alpha/2)$

$$CI: \bar{x} - z_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{s}{\sqrt{n}}$$

SAMPLE STANDARD DEVIATION

SAMPLE STANDARD DEVIATION

5. DIFFERENCE OF TWO MEAN, FIND CI

5.1 — KNOWN VARIANCE

σ_1^2 AND σ_2^2 KNOWN

qnorm(1−α/2)

$$CI: \left((\bar{x}-\bar{y}) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}\ ;\ (\bar{x}-\bar{y}) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}\right)$$

5.2 — SAME AND UNKNOWN VARIANCE

← $\sigma_1^2 = \sigma_2^2$ AND UNKNOWN

qT(1−α/2; n₁+n₂−2)

$$CI: \left((\bar{x}-\bar{y}) - t_{\alpha/2}^{n_1+n_2-2} S_c\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\ ;\ (\bar{x}-\bar{y}) + t_{\alpha/2}^{n_1+n_2-2} S_c\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right)$$

$$S_c = \sqrt{\frac{S_1^2(n_1-1)+S_2^2(n_2-1)}{n_1+n_2-2}} \qquad S_1^2 = \frac{\sum(x_i-\bar{x})^2}{n_1-1} \qquad S_2^2 = \frac{\sum(y_i-\bar{y})^2}{n_2-1}$$

5.3 — INDIPENDENT POPULATION W/ LARGE SAMPLE

- VARIANCE COULD BE DIFFERENT OR NOT

ASYMPTOTIC CONFIDENCE INTERVAL

$$ACI: \left((\bar{x}-\bar{y}) - z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}\ ;\ (\bar{x}-\bar{y}) + z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}\right)$$

BINOMIAL (PAG 29)

2. NORMAL POPULATION, POP. UNKNOWN VARIANCE AND FIND CI FOR THE POP. MEAN

- WITH UNKNOWN POP. VARIANCE THE CI IS WIDER THAN THE ONE WITH VARIANCE

- $t_{n-1} = qT(1-\alpha/2; n-1)$

$$CI: \bar{x} - t_{n-1}\cdot\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1}\cdot\frac{s}{\sqrt{n}}$$

SAMPLE STANDARD DEVIATION

SAMPLE STANDARD DEVIATION

- $\text{LENGTH}(CI) = L_2 - L_1 = 2t_{n-1}\frac{s}{\sqrt{n}}$

4. BERNULLI POPULATION, LARGE SAMPLE AND FIND CI FOR THE POP. MEAN

APPLY IF

- $n \geq 30$ W/ VARIANCE KNOWN
- $n \geq 30$ W/ VARIANCE UNKNOWN

] → CLT (CENTRAL LIMIT THEOREM)

- $z_{\alpha/2} = \text{qnorm}(1-\alpha/2)$

$$CI: \bar{x} - z_{\alpha/2}\frac{\sqrt{\bar{x}(1-\bar{x})}}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sqrt{\bar{x}(1-\bar{x})}}{\sqrt{n}}$$

6. CI FOR THE VARIANCE

FOR ONE AT TIME

$$CI: \left((n-1)\frac{S^2}{\chi^2_{\alpha/2}}\ ;\ (n-1)\frac{S^2}{\chi^2_{1-\alpha/2}}\right)$$

$$PQ = \frac{\sum(x_i-\mu)^2}{\sigma^2} \sim \chi_n^2$$

$$\chi^2_{\alpha/2} = \text{qchisq}(1-\alpha/2; N-1)$$

ALWAYS THE COMPLEMENTARY

7. CI FOR THE RATIO OF 2 VARIANCE

$$CI: \left(\frac{S_y^2}{S_x^2}F_{1-\alpha/2} < \frac{\sigma_y^2}{\sigma_x^2} < \frac{S_y^2}{S_x^2}F_{\alpha/2}\right)$$

QUANTILE

QUANTILE

TO KNOW:

- BERNULLI $S^2 = P(1-P)$; $S = \sqrt{P(1-P)}$
- $\bar{X} \sim N(\mu_1, \sigma_1^2)$; $\bar{Y} \sim N(\mu_2; \sigma_2^2) \longrightarrow (\bar{X}-\bar{Y}) \sim N(\mu_1-\mu_2; \sigma_1^2+\sigma_2^2)$ (ALWAYS SUM)
- $P(|T| > 1.428) = 2(1-\Phi(1.428))$
- $qT(1-\alpha; n-1) = -qt(\alpha; n-1)$
- SERIE: $\sum_{k=1}^{+\infty}\left(\frac{1}{2^\alpha}\right)^k = c\,\frac{\frac{1}{2^\alpha}}{1-\frac{1}{2^\alpha}}$
- AN ESTIMATOR IS UNBIASED: (IF) $E(T) - \theta = 0$ (ESTIMATOR)
- BIASED: $E(T) \neq \theta$

DIFFERENCE OF TWO SAMPLE MEAN

IN GENERAL, WITH TWO DIFFERENT DISTRIBUTION

- $E(\bar{x}-\bar{y}) = \mu_x - \mu_y$
- $VAR(\bar{x}-\bar{y}) = \frac{\sigma_x^2}{n_1} + \frac{\sigma_y^2}{n_2}$

NORMAL: $x \sim N(\mu_x, \sigma_x^2)$; $y \sim N(\mu_y, \sigma_y^2)$

$$\Rightarrow (\bar{x}-\bar{y}) \sim N\left(\mu_x-\mu_y; \frac{\sigma_x^2}{n_1}+\frac{\sigma_y^2}{n_2}\right)$$

BERNULLI WITH LARGE SAMPLE: $x \sim BIN(1, p_x)$; $y \sim BIN(1, p_y)$

$$\Rightarrow (\bar{x}-\bar{y}) \sim N\left(p_x-p_y; \frac{p_x(1-p_x)}{n_1}+\frac{p_y(1-p_y)}{n_2}\right)$$

- Check n
  - n < 30
    - σ^2 known
      - r ~ N(μ, σ^2) — Already Normal
      - r ~ unk — NO.
    - σ^2 unknown
      - r ~ N(μ, σ^2) — T-student, S as standard deviation
      - r ~ unk — NO.
  - n > 30
    - 30 < n < 50
      - σ^2 known
        - r ~ N(μ, σ^2) — Already Normal
        - r ~ unk — CLT (normalize), σ as standard deviation
      - σ^2 unknown
        - r ~ N(μ, σ^2) — T-student, S as standard deviation
        - r ~ unk — NO.
    - n > 50
      - σ^2 known
        - r ~ N(μ, σ^2) — Already Normal
        - r ~ unk — CLT (normalize)
      - σ^2 unknown
        - r ~ N(μ, σ^2) — T-student, S as standard deviation
        - r ~ unk — CLT (normalize), S as standard deviation

HYPOTESIS TESTING

TYPE I ERROR $\rightarrow P(\text{REJ. } H_0 \mid H_0 \text{ TRUE}) = \alpha$

TYPE II ERROR $\rightarrow P(\text{KEEP } H_0 \mid H_0 \text{ FALSE}) = \beta$

POWER: $1-\beta$

P-VALUE: $\gamma \rightarrow$ A MORE EXTREME VALUE THAN T_{OBS}

HT ON μ, $X \sim N(\mu, \sigma^2)$, σ^2 KNOWN

HT ON μ, $X \sim$ UNKNOWN, LARGE n

$n \geq 30$

HP ON P, $X \sim BERN(P)$, LARGE n

$n \geq 30$

$\rightarrow \sigma = \sqrt{P_0(1-P_0)}$

AKA μ in BERNULLI

- RT TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu = \mu_1 > \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$; $z_\alpha = \text{QNORM}(1-\alpha)$; $\gamma = 1 - \text{PNORM}(z_{OBS})$
  - $z < z_\alpha \rightarrow$ KEEP H_0
  - $z > z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu = \mu_1 < \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$; $z_\alpha = \text{QNORM}(1-\alpha)$; $\gamma = \text{PNORM}(z_{OBS})$
  - $z > -z_\alpha \rightarrow$ KEEP H_0
  - $z < -z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$; $z_{\alpha/2} = \text{QNORM}(1-\alpha/2)$; $\gamma = 2(1-\text{PNORM}(|z_{OBS}|))$
  - $-z_{\alpha/2} < z < z_{\alpha/2} \rightarrow$ KEEP H_0
  - $z < -z_{\alpha/2} \vee z > z_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0

HT ON μ, $X \sim N(\mu, \sigma^2)$, σ^2 UNKNOWN

- RT TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu = \mu_1 > \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $T = \dfrac{\bar{x}-\mu_0}{S/\sqrt{n}}$; $t_\alpha = qT(1-\alpha; n-1)$; $\gamma = 1 - PT(T_{OBS}; n-1)$
  - $T_{OBS} < t_\alpha \rightarrow$ KEEP H_0
  - $T_{OBS} > t_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu = \mu_1 < \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $T = \dfrac{\bar{x}-\mu_0}{S/\sqrt{n}}$; $t_\alpha = -qT(1-\alpha; n-1)$; $\gamma = PT(T_{OBS}; n-1)$
  - $T_{OBS} > -t_\alpha \rightarrow$ KEEP H_0
  - $T_{OBS} < -t_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $T = \dfrac{\bar{x}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$; $t_{\alpha/2} = qT(1-\alpha/2; n-1)$; $\gamma = 2(1-PT(|T_{OBS}|; n-1))$
  - $-t_{\alpha/2} < T_{OBS} < t_{\alpha/2} \rightarrow$ KEEP H_0
  - $T_{OBS} < -t_{\alpha/2} \vee T_{OBS} > t_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0

HT ON $\mu_1 - \mu_2$, $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma^2)$, $X \perp\!\!\!\perp Y$, σ_1^2 AND σ_2^2 KNOWN

$(\bar{x}-\bar{y}) \sim N(\mu_1 - \mu_2; \sigma_1^2/n_1 + \sigma_2^2/n_2)$ IF $\mu_1 - \mu_2 = d$

- RT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 > \mu_2 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{(\bar{x}-\bar{y}) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ (or d) $\rightarrow$ ⚠ NO $\sqrt{n}$; $z_\alpha = \text{QNORM}(1-\alpha)$; $\gamma = 1 - \text{PNORM}(z_{OBS})$, $\hookrightarrow \Phi(z_{OBS})$
  - $z < z_\alpha \rightarrow$ KEEP H_0
  - $z > z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 < \mu_2 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{(\bar{x}-\bar{y}) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ (or d) $\rightarrow$ ⚠ NO $\sqrt{n}$; $-z_\alpha = -\text{QNORM}(1-\alpha)$; $\gamma = \text{PNORM}(z_{OBS})$, $\hookrightarrow \Phi(z_{OBS})$
  - $z > -z_\alpha \rightarrow$ KEEP H_0
  - $z < -z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 \neq \mu_2 \end{cases} \rightarrow$ STANDARDIZE $\bar{x}$: $z = \dfrac{(\bar{x}-\bar{y}) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ (or d) $\rightarrow$ ⚠ NO $\sqrt{n}$; $z_{\alpha/2} = \text{QNORM}(1-\alpha/2)$; $\gamma = 2(1-\text{PNORM}(|z_{OBS}|))$, $\hookrightarrow \Phi(|z_{OBS}|)$
  - $-z_{\alpha/2} < z < z_{\alpha/2} \rightarrow$ KEEP H_0
  - $z < -z_{\alpha/2} \vee z > z_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0

HT ON $\mu_1 - \mu_2$, $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$; $X \perp\!\!\!\perp Y$ $\sigma_1^2 = \sigma_2^2$ UNKNOWN

HT ON $\mu_1 - \mu_2$, $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$; $X \perp\!\!\!\perp Y$ $\sigma_1^2 \neq \sigma_2^2$

$\rightarrow$ IDENTICAL BUT THE TEST STATISTIC IS $T = \dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

$T = \dfrac{\bar{x}-\bar{y}}{S_{xx}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$ $\qquad S_{xx} = \sqrt{\dfrac{(n_1-1)^2 S_1^2 + (n_2-1)^2 S_2^2}{n_1+n_2-2}}$

- RT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 > \mu_2 \end{cases} \rightarrow$ STANDARDIZE T_{OBS}: $t_\alpha = qT(1-\alpha; n_1+n_2-2)$; $\gamma = 1 - PT(T_{OBS}; n_1+n_2-2)$
  - $T_{OBS} < t_\alpha \rightarrow$ KEEP H_0
  - $T_{OBS} > t_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 < \mu_2 \end{cases} \rightarrow$ STANDARDIZE T_{OBS}: $t_\alpha = qT(1-\alpha; n_1+n_2-2)$; $\gamma = PT(T_{OBS}; n_1+n_2-2)$
  - $T_{OBS} > -t_\alpha \rightarrow$ KEEP H_0
  - $T_{OBS} < -t_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 \neq \mu_2 \end{cases} \rightarrow$ STANDARDIZE T_{OBS}: $-t_{\alpha/2} = -qT(1-\alpha/2; n_1+n_2-2)$; $\gamma = 2(1-PT(|T_{OBS}|; n_1+n_2-2))$
  - $-t_{\alpha/2} < T_{OBS} < t_{\alpha/2} \rightarrow$ KEEP H_0
  - $T_{OBS} < -t_{\alpha/2} \vee T_{OBS} > t_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0

HP ON $\mu_1 - \mu_2$; $X \sim$ UNK, $Y \sim$ UNK; n_1 AND n_2 LARGE

$n \geq 30$

$(\bar{x}-\bar{y}) \sim N(\mu_1 - \mu_2; \sigma_1^2/n_1 + \sigma_2^2/n_2)$

- RT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 > \mu_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$; $z_\alpha = \text{QNORM}(1-\alpha)$; $\gamma = 1 - \text{PNORM}(z_{OBS})$
  - $z_{OBS} < z_\alpha \rightarrow$ KEEP H_0
  - $z_{OBS} > z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 < \mu_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$; $-z_\alpha = -\text{QNORM}(1-\alpha)$; $\gamma = \text{PNORM}(z_{OBS})$
  - $z_{OBS} > -z_{\alpha/2} \rightarrow$ KEEP H_0
  - $z_{OBS} < -z_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 \neq \mu_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$; $z_{\alpha/2} = \text{QNORM}(1-\alpha/2)$; $\gamma = 2(1-\text{QNORM}(|z_{OBS}|))$
  - $-z_{\alpha/2} < z_{OBS} < z_{\alpha/2} \rightarrow$ KEEP H_0
  - $z_{OBS} < -z_{\alpha/2} \vee z_{OBS} > z_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma > \alpha \rightarrow$ REJECT H_0

HP ON $P_1 - P_2$; $X \sim BERN(P_1)$; $Y \sim BERN(P_2)$; LARGE SAMLE

$n \geq 30$

$\bar{x}-\bar{y} \sim N(\mu_1 - \mu_2; \sigma_1^2/n_1 + \sigma_2^2/n_2)$

- RT TAIL
  $\begin{cases} H_0: P_1 = P_2 \\ H_1: P_1 > P_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{P(1-P)(1/n_1 + 1/n_2)}}$; $z_\alpha = \text{QNORM}(1-\alpha)$; $\gamma = 1 - \text{PNORM}(z_{OBS})$
  - $z_{OBS} < z_\alpha \rightarrow$ KEEP H_0
  - $z_{OBS} > z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- LT TAIL
  $\begin{cases} H_0: P_1 = P_2 \\ H_1: P_1 < P_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{P(1-P)(1/n_1 + 1/n_2)}}$; $-z_\alpha = -\text{QNORM}(1-\alpha)$; $\gamma = \text{PNORM}(z_{OBS})$
  - $z_{OBS} > -z_\alpha \rightarrow$ KEEP H_0
  - $z_{OBS} < -z_\alpha \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0
- TWO SIDE TAIL
  $\begin{cases} H_0: P_1 = P_2 \\ H_1: P_1 \neq P_2 \end{cases} \rightarrow$ STANDARIZE z: $z = \dfrac{\bar{x}-\bar{y}}{\sqrt{P(1-P)(1/n_1 + 1/n_2)}}$; $z_{\alpha/2} = \text{QNORM}(1-\alpha/2)$; $\gamma = 2(1-\text{PNORM}(|z_{OBS}|))$
  - $-z_{\alpha/2} < z_{OBS} < z_{\alpha/2} \rightarrow$ KEEP H_0
  - $z_{OBS} < -z_{\alpha/2} \vee z_{OBS} > z_{\alpha/2} \rightarrow$ REJECT H_0
  - $\gamma < \alpha \rightarrow$ REJECT H_0

HT ON σ^2, $X \sim N(\mu, \sigma^2)$, μ KNOWN

?

HT ON σ^2, $X \sim N(\mu, \sigma^2)$, μ UNKNOWN

TEST STAT: $\dfrac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}$

- $H_1: \sigma^2 > \sigma_0^2$ IF $n-1/\sigma_0^2 \cdot S^2 > \text{QCHISQ}(1-\alpha; n-1)$
- $H_1: \sigma^2 < \sigma_0^2$ IF $n-1/\sigma_0^2 \cdot S^2 < \text{QCHISQ}(\alpha; n-1)$
- $H_1: \sigma^2 \neq \sigma_0^2$ IF $n-1/\sigma_0^2 \cdot S^2 > \text{QCHISQ}(1-\alpha/2; n-1)$, $< \text{QCHISQ}(\alpha/2; n-1)$

HT ON σ_1^2/σ_2^2 $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim$??